PARASIGHT: An Ai-Based Microscopic Image Analysis System For Coproparasitological Parasite Detection

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RESEARCH PLAN

Rationale

Intestinal parasitic infections continue to pose a significant public health concern, particularly in settings where sanitation, access to diagnostic laboratories, and trained personnel are limited. Accurate identification of parasites in stool specimens is essential for proper diagnosis, treatment, surveillance, and monitoring of infection control efforts. Conventional coproparasitological examination remains a widely used and low-cost diagnostic approach; however, it depends heavily on manual microscopic analysis, which is labor-intensive, time-consuming, and prone to inter-observer variability (Dąbrowska et al., 2024).

Manual examination can be especially challenging when parasite structures are morphologically similar to non-parasitic materials, present in low concentrations, or obscured by fecal debris and other microscopic artifacts. These limitations may contribute to missed detections, inconsistent results, and delayed diagnosis. Because microscopic examination requires skilled personnel and careful visual inspection, laboratories may face difficulties maintaining consistent diagnostic quality, particularly when processing a high volume of specimens (Dąbrowska et al., 2024).

Recent advances in artificial intelligence (AI) and computer vision offer a promising opportunity to improve microscopic image analysis. Deep learning models can be trained to recognize visual patterns associated with parasites, potentially supporting laboratory personnel by automatically detecting and classifying suspected parasitic structures in digital microscopic images. Mathison et al. (2020) demonstrated the feasibility of using a deep convolutional neural network to detect intestinal protozoa in

trichrome-stained stool specimens, showing that AI-based image analysis can complement conventional microscopy rather than replace it.

The system aims to automate the detection and identification of intestinal parasites from microscopic stool images. Its performance will be evaluated using precision, recall, F1-score, classification accuracy, and testing under varying image conditions. Expert-verified annotations will serve as reference data to assess the reliability and consistency of the system's outputs.

Ultimately, this study seeks to demonstrate the potential of artificial intelligence in supporting more organized, consistent, and efficient microscopic parasite analysis. PARASIGHT is intended as a research-support and decision-support tool that can assist laboratory personnel; it is not intended to replace professional laboratory examination or expert clinical interpretation.

Objectives

The study aims to develop PARASIGHT, an AI-based microscopic image analysis system for coproparasitological examination. Specifically, it seeks to evaluate the system's detection performance, classification accuracy, response to varying microscopic image conditions, analysis time, and common detection and classification errors. Specifically, this study seeks:

1. To Collect and organize microscopic fecal images containing selected intestinal parasites and establish expert-verified reference annotations.

2. To Design an AI-based detection model and classification system for identifying selected intestinal parasite types in microscopic fecal images.

3. To Develop an offline website integrating the AI model for microscopic image uploading, automated parasite detection, classification, and presentation of results.

4. To determine the detection performance of PARASIGHT in identifying selected intestinal parasites in microscopic fecal images in terms of:
   1.1. Precision;
   1.2. Recall;
   1.3. F1-score; and
   1.4 Classification accuracy

5. To assess the effects of variations in microscopic image conditions, particularly image clarity, illumination, and the presence of fecal debris, on the detection performance of PARASIGHT.

6. To compare the time required by PARASIGHT to analyze microscopic fecal images with the time required for manual analysis of the same images.

7. To identify common detection and classification errors encountered by PARASIGHT and determine the limitations these errors reveal regarding the system's performance.

Hypotheses

Null Hypothesis (H_0)

There is no significant difference in PARASIGHT's detection and classification performance under varying microscopic image conditions, nor in the analysis time between PARASIGHT and manual microscopic analysis.

Alternative Hypothesis (H_1)

There is a significant difference in PARASIGHT's detection and classification performance under varying microscopic image conditions, and in the analysis time between PARASIGHT and manual microscopic analysis.

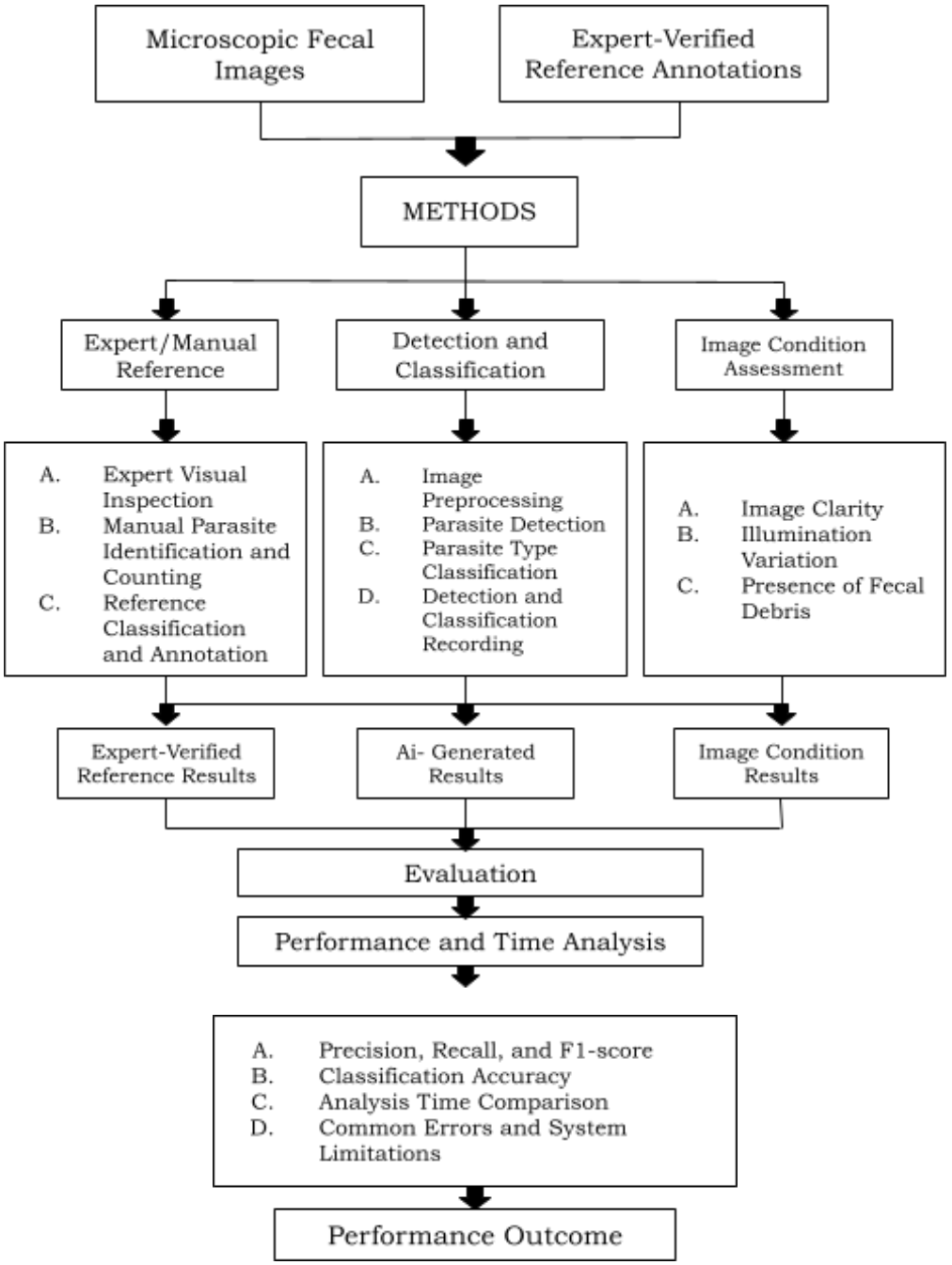
Goals/Expected Outcomes

The project is expected to develop a functional version of PARASIGHT with offline functionality, an AI-based microscopic image analysis system designed to detect and classify selected intestinal parasites in microscopic fecal images (Yarahmadi, Ahmadpour, Moradi, & Samadzadehaghdam, 2025). The website will allow users to upload images, process them using a trained AI model, and view detection results without requiring an active internet connection, provided that the application and its required components are installed locally.

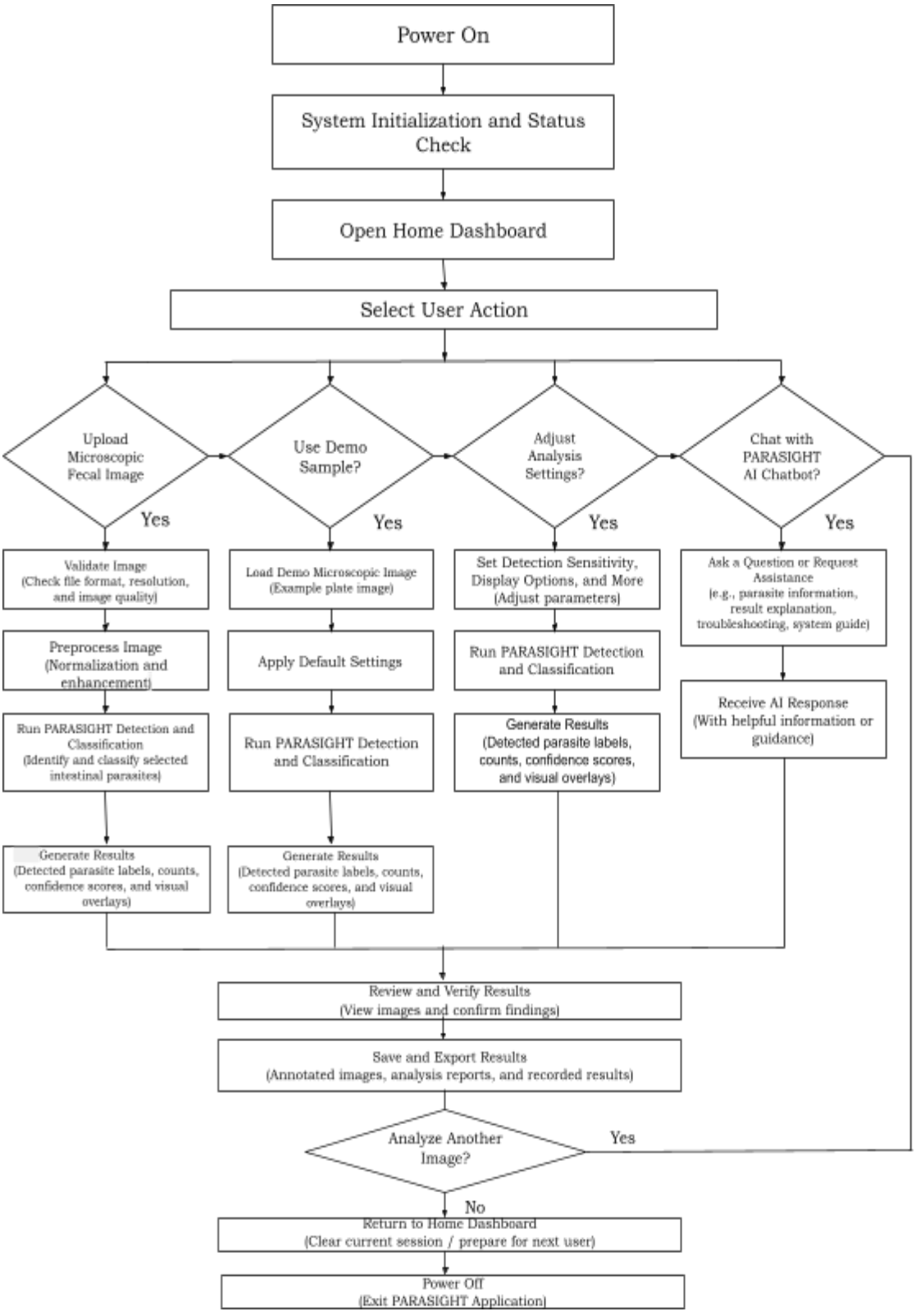
The project will also produce a prepared and documented dataset inventory containing information about the image sources, parasite categories, annotations, and data preparation procedures. The system's performance will be evaluated using precision, recall, F1-score, and classification accuracy (Sokolova & Lapalme, 2016). Additionally, the study will assess the effects of varying image conditions on detection performance (Anorboev et al., 2024), compare the system's analysis time with manual image analysis, and document common errors and technical limitations.

Furthermore, the project will generate a comprehensive technical evaluation report detailing the system's detection and classification performance, offline functionality, processing time, and identified limitations. The final evaluation will provide a basis for determining the system's effectiveness and identifying areas for improvement in AI-assisted coproparasitological image analysis.

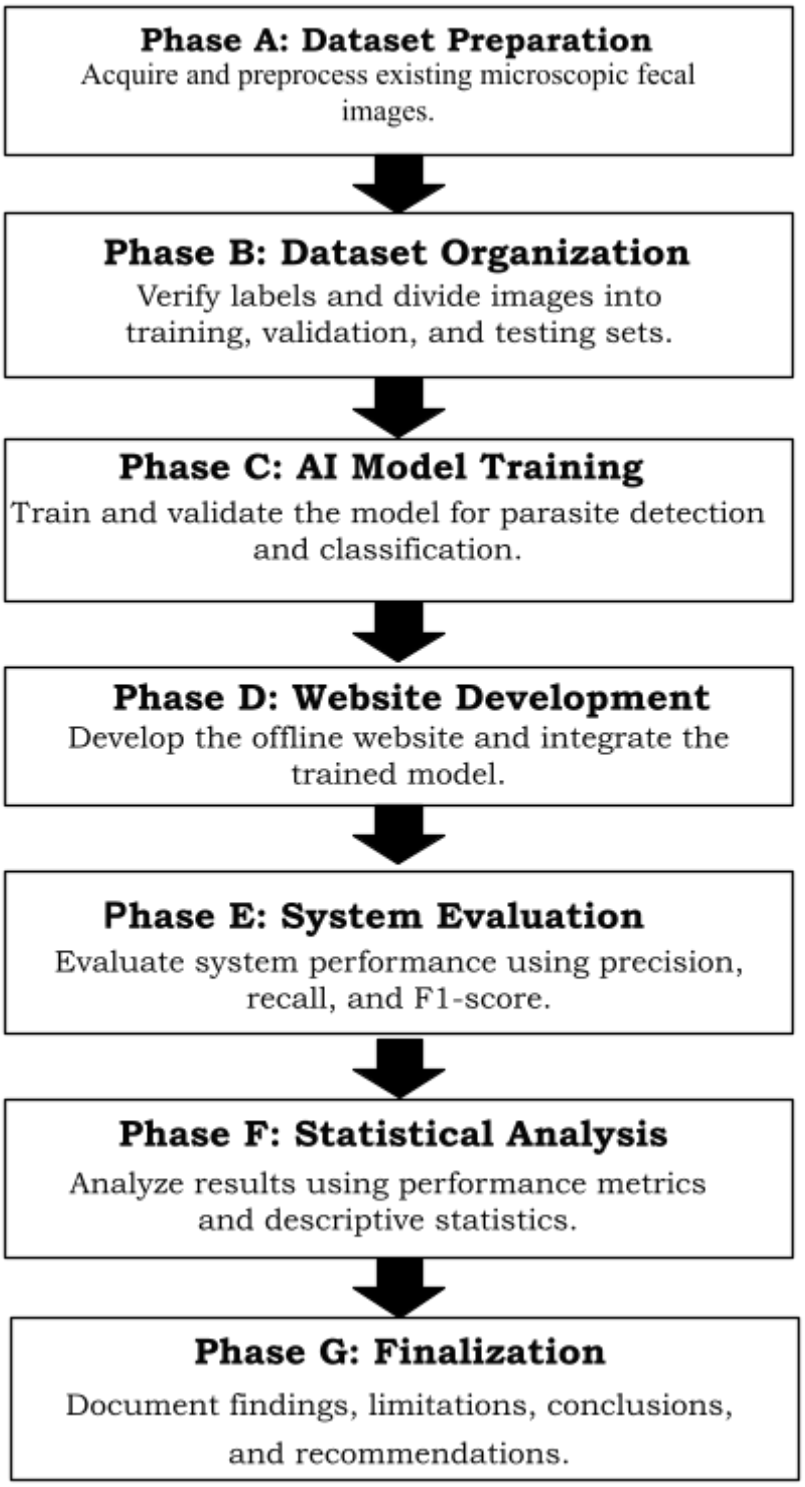
Methodological Framework



System flow


Power On
System Initialization and Status Check
Open Home Dashboard
Select User Action
Upload Microscopic Fecal Image
Use Demo Sample?
Adjust Analysis Settings?
Chat with PARASIGHT AI Chatbot?
Yes
Yes
Yes
Yes
Validate Image (Check file format, resolution, and image quality)
Load Demo Microscopic Image (Example plate image)
Set Detection Sensitivity, Display Options, and More (Adjust parameters)
Ask a Question or Request Assistance (e.g., parasite information, result explanation, troubleshooting, system guide)
Preprocess Image (Normalization and enhancement)
Apply Default Settings
Run PARASIGHT Detection and Classification
Receive AI Response (With helpful information or guidance)
Run PARASIGHT Detection and Classification (Identify and classify selected intestinal parasites)
Run PARASIGHT Detection and Classification
Generate Results (Detected parasite labels, counts, confidence scores, and visual overlays)
Generate Results (Detected parasite labels, counts, confidence scores, and visual overlays)
Generate Results (Detected parasite labels, counts, confidence scores, and visual overlays)
Review and Verify Results (View images and confirm findings)
Save and Export Results (Annotated images, analysis reports, and recorded results)
Analyze Another Image?
Yes
No
Return to Home Dashboard (Clear current session / prepare for next user)
Power Off (Exit PARASIGHT Application)

Procedures


Phase A: Dataset Preparation
Acquire and preprocess existing microscopic fecal images.
Phase B: Dataset Organization
Verify labels and divide images into training, validation, and testing sets.
Phase C: AI Model Training
Train and validate the model for parasite detection and classification.
Phase D: Website Development
Develop the offline website and integrate the trained model.
Phase E: System Evaluation
Evaluate system performance using precision, recall, and F1-score.
Phase F: Statistical Analysis
Analyze results using performance metrics and descriptive statistics.
Phase G: Finalization
Document findings, limitations, conclusions, and recommendations.

Phase A: Identification and Preparation of Existing Microscopic Image Data

The researchers will acquire existing microscopic fecal image datasets from Mendeley Data and the LIDS-UNICAMP GitHub repository. The datasets will be examined based on image quality, relevance, parasite categories, and available annotations. The selected images will be organized according to eight proposed parasite categories: Ascaris lumbricoides, Trichuris trichiura, Ancylostoma (Hookworm), Enterobius vermicularis, Taenia spp., Giardia intestinalis, Entamoeba histolytica, and Entamoeba coli.

Phase B: Image Label Verification and Dataset Organization

The researchers will verify the parasite labels and available annotations to ensure their accuracy and compatibility with the YOLO algorithm. The images will be checked for duplicates, inconsistent labels, and unusable files. The annotations will be converted into the required YOLO format, including bounding boxes around identified parasite structures. The dataset will then be divided into 70% for training and 30% for testing, with a validation subset obtained from the training data.

Phase C: Development and Training of the AI-Based Parasite Detection

The researchers will use the YOLO algorithm to develop an AI-based model for detecting and classifying the selected intestinal parasites. The prepared images will undergo resizing, normalization, and other necessary preprocessing procedures before training. The model will be trained using the training dataset and monitored through the validation subset. After training, the model will be evaluated, adjusted when necessary, and saved for local deployment to support offline image analysis.

Phase D: Website Development and AI model Integration

The researchers will develop the PARASIGHT website with essential features, including image uploading, processing, and results display. The trained YOLO model will be integrated to analyze uploaded microscopic fecal images and generate predicted parasite categories, annotated images with bounding boxes, and corresponding confidence scores. The website and its required components will also be configured for offline accessibility, allowing image analysis without an active internet connection.

Phase E: System Testing and Performance Evaluation

The researchers will conduct functional testing to verify the website's image uploading, processing, results display, and offline capabilities. The reserved testing dataset will then be used to evaluate the model's detection and classification performance based on expert-verified reference annotations. Precision, recall, F1-score, and classification accuracy will be determined. The researchers will also assess performance under varying image clarity, illumination, and fecal debris conditions, compare the system's analysis time with manual image analysis, and document common detection errors and limitations.

Phase F: Statistical Evaluation of Detection Accuracy and System Efficiency

The researchers will organize and analyze the collected results using appropriate statistical methods. A confusion matrix will be prepared to examine correct and incorrect classifications across the selected parasite categories. Precision, recall, F1-score, and classification accuracy will be calculated, while descriptive statistics will be used to summarize

performance under different image conditions and compare the analysis time of PARASIGHT with manual analysis.

Phase G: Finalization and Documentation

The researchers will consolidate and interpret the findings to determine whether PARASIGHT meets its intended objectives. The final documentation will include the website's functions, detection and classification performance, system efficiency, observed limitations, and potential improvements. Conclusions and recommendations will then be formulated based on the results of the study.

Risk and Safety

The PARASIGHT system may encounter several potential hazards during development and testing. Data integrity risks include incorrect dataset labels, which may result in unreliable parasite detection. Data availability risks involve insufficient or imbalanced datasets that may cause biased predictions. Data leakage may occur when duplicate images appear in training and testing sets, leading to misleading results (Ross, 2012).

Image quality hazards, such as blurry images, poor illumination, and unclear microscopic details, may cause missed parasite detection. AI classification errors may occur when fecal debris is mistaken for parasites or parasite types are incorrectly identified.

Data licensing hazards involve unauthorized use of dataset images, while privacy risks include the exposure of sensitive information. Cybersecurity threats, such as malicious uploads and unauthorized website access, may compromise research data. Software and hardware failures may interrupt image processing or cause data loss (Marron, 2024).

Lastly, result interpretation risks arise when users mistake AI predictions for confirmed diagnoses. File management hazards include accidental deletion or corruption of datasets, model files, and research results.

To minimize these risks, the researchers should use verified datasets, test images under different conditions, protect sensitive information, secure the website, require expert verification of results, and maintain organized backups (Ross, 2012).

Data Analysis

The data from the evaluation of PARASIGHT are analyzed using descriptive and inferential statistical methods. Descriptive statistics, including frequency, percentage, mean, standard deviation, and range, are used to summarize the system's detection and classification performance, analysis time, and error patterns (Field, 2018).

For detection evaluation, each prediction is labeled as a true positive (TP) if PARASIGHT detects a parasite object that matches an expert-annotated parasite within the accepted overlap threshold; a false positive (FP) if it detects a non-parasitic object or an incorrect parasite object; and a false negative (FN) if it fails to detect an expert-annotated parasite. These counts are used to calculate precision, recall, and F1-score (Sokolova & Lapalme, 2009).

Detection Performance

The detection performance of PARASIGHT in identifying selected intestinal parasites is evaluated using precision, recall, and F1-score. Precision measures the proportion of predicted parasites that are correctly identified, recall measures the proportion of actual parasites correctly

detected, and the F1-score provides a balanced measure of the two metrics (Sokolova & Lapalme, 2009; Powers, 2011).

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1\text{-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Classification Accuracy

PARASIGHT's classification accuracy is evaluated by comparing its predicted parasite types with expert-verified annotations. A confusion matrix is used to identify correct and incorrect classifications, from which overall and per-class accuracy are calculated (Stehman, 1997).

$$Overall\ Accuracy = \frac{Number\ of\ Correct\ Classifications}{Total\ Number\ of\ Classified\ Parasite\ Objects} \times 100$$

$$Class\ Accuracy = \frac{Correct\ Predictions\ for\ Parasite\ Type}{Total\ Reference\ Annotations\ for\ Parasite\ Type} \times 100$$

Analysis Time Comparison

The analysis time of PARASIGHT is compared with manual examination performed by trained laboratory personnel using the same microscopic fecal images. Descriptive statistics, including mean, standard deviation, median, minimum, and maximum analysis time, are calculated for both methods (Field, 2018).

The percentage reduction in analysis time is also determined using the following formula:

$$Time\ Reduction\ (\%) = \frac{Manual\ Time - PARASIGHT\ Time}{Manual\ Time} \times 100$$

Detection And Classification Errors

Common detection and classification errors are identified by reviewing all false positives, false negatives, and misclassified parasite detections. Each error is categorized according to its probable cause, such as missed parasites due to low parasite density or poor image quality, false detections caused by debris or other artifacts, and misclassification due to morphological similarity, partial visibility, or overlapping objects (Sokolova & Lapalme, 2009; Stehman, 1997).

The frequency and percentage of each error type are calculated using the following formulas:

$$Error\ Frequency\ (\%) = \frac{Number\ of\ Errors\ of\ a\ Given\ Type}{Total\ Number\ of\ Errors} \times 100$$

The most frequent error categories are interpreted as major limitations of PARASIGHT and are used to recommend possible improvements in image preprocessing, parasite detection, and classification.

Ethical Consideration

The study will observe ethical principles to protect the privacy, safety, and rights of individuals and communities. Informed consent will be obtained when human participants or personal data are involved, ensuring voluntary participation and the right to withdraw (HHS, 2017; WHO, 2021).

Privacy and confidentiality will be maintained by securing data, anonymizing personal information, and using only necessary information. Researchers will also ensure proper handling and acknowledgment of microscopic images and their sources (NCBI, 2022; ICMJE, 2025).

Since PARASIGHT is an AI-based system, possible errors in data recording, image classification, and system performance will be reviewed and documented. Results will be reported honestly without fabrication or manipulation. The system will serve only as an assistive research tool and will not replace professional laboratory examination or medical diagnosis (WHO, 2021).

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